1 point series - parallel testing board with a exhaust fan

PROJECT REPORT

Submitted by

BADAL SINGH

Regd.No - 230301120099

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF ENGINEERING AND TECHNOLOGY BHUBANESWAR CAMPUS

CENTURION UNIVERSITY OF TECHNOLOGY AND MANAGEMENT ODISHA

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DEPARTMENT OF COMPUTER SCIENCE

SCHOOL OF ENGINEERING AND TECHNOLOGY

BHUBANESWAR CAMPUS

BONAFIDE CERTIFICATE

Certified that this project report of "**1 POINT SERIES – PARALLAL TESTING BOARD WITH A EXHAUST FAN**" is the Bonafide work of **Badal Singh** who carried out the project work under my supervision. This is to further certify to the best of my knowledge, that this project has not been carried out earlier in this institute and the university.

SIGNATURE

(Mr. Amit Kumar Sahoo)
Asst. Professor of ECE, SoET

Certified that the above-mentioned project has been duly carried out as per the norms of the college and statutes of the university.

SIGNATURE

(Dr. Raj Kumar Mohanta)

HEAD OF THE DEPARTMENT / DEAN OF THE SCHOOL

Professor of Computer Scienceand Engineering

DEPARTMENT SEAL

DECLARATION

I hereby declare that the project entitled "**1 point series - parallel testing board with a exhaust fan**" submitted for the "Minor Project" of 5th semester B. Tech in Computer Science and Engineering is my original work and the project has not formed the basis for the award of any Degree / Diploma or any other similar titles in any other University / Institute.

Name of the student: Badal Singh

Signature of the student:

Registration no: 230301120099

Place : Bhubaneswar, CUTM

Date:

ACKNOWLEDGEMENTS

I wish to express my profound and sincere gratitude **to Dr. Amit Kumar Sahoo** Department of Associate Professor of SoET, Bhubaneswar Campus, who guided me into the intricacies of this project nonchalantly with matchless magnanimity.

I thank **Prof. Raj Kumar Mohanta**, Head of the Dept. of Department of Computer Science and Engineering, SoET, Bhubaneswar Campus and **Prof. Sujata Chakrabarty**, Dean, School of Engineering and Technology, Bhubaneswar Campus for extending their support during Course of this investigation.

I would be failing in my duty if I don't acknowledge the cooperation rendered during various stages of image interpretation **by**

I am highly grateful to **Dr. Amit Kumar Sahoo** who evinced keen interest and invaluable support in the progress and successful completion of my project work.

I am indebted **to Dr. Amit Kumar Sahoo** for their constant encouragement, co- operation and help. Words of gratitude are not enough to describe the accommodation and fortitude which they have shown throughout my endeavor.

Name of the student: Badal Singh

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Place : Bhubaneswar, CUTM

Date:

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CHAPTER1 - Abstract

The project **“1 Point Series–Parallel Testing Board with Exhaust Fan”** is designed to test various electrical appliances safely in both series and parallel modes. The main objective of this project is to provide a controlled testing setup where electrical devices can be tested at low voltage (series connection) and full voltage (parallel connection) as required.

The setup includes a fuse for protection, a bulb as a test load**,** a 3-pin socket for appliance connection**,** a control switch for mode selection, and an exhaust fan for cooling and ventilation. The series connection allows testing at reduced voltage to ensure the safety of the appliance and the user, while the parallel connection allows normal operation at full supply voltage.

This project helps students and technicians understand the practical behavior of series and parallel circuits**,** load characteristics**,** and the importance of safety measures in electrical testing. The exhaust fan ensures continuous ventilation, preventing overheating of components.

In conclusion, this project demonstrates the efficient use of basic electrical components to build a safe, functional, and educational testing board for small electrical appliances

CHAPTER 2 -Introduction

1. Importance of Electrical Testing Boards

Electrical testing boards are fundamental tools in the field of electrical engineering and education. They serve as a vital bridge between theoretical knowledge and practical, hands-on application. By providing a controlled, safe, and organized environment, testing boards allow students, technicians, and engineers to construct, analyze, and troubleshoot circuits without the hazards associated with open or improvised wiring. They are essential for verifying circuit laws, understanding component behavior, and developing practical skills in a systematic manner, forming the backbone of any electrical laboratory or workshop.

2. Need for Series and Parallel Circuit Testing

Understanding **series and parallel circuits** is the non-negotiable foundation of all electrical and electronic principles. These two configurations are the basic building blocks for every complex circuit, from simple household wiring to sophisticated microprocessors.

- **Series circuits** demonstrate key concepts like voltage division and current consistency, and show how a break anywhere in the path stops the entire circuit.
- **Parallel circuits** illustrate principles like current division and constant voltage, which is the basis for wiring buildings (e.g., all outlets receive the same voltage).

The ability to practically test and visually compare these two setups is crucial. It helps solidify abstract concepts, such as why bulbs in a series circuit dim as more are added, while bulbs in parallel maintain their brightness.

3. Objective of the 1 Point Series–Parallel Testing Board

The primary objective of this project is to design and build a **single, compact, and cost-effective unit** that can instantly switch a single test point (or load) between a series and a parallel configuration.

This "1 point" design eliminates the need for complex re-wiring, allowing a user to directly and immediately observe the different behaviors of the two circuit types. A secondary objective is to integrate an **exhaust fan** for thermal management. This ensures the board can be operated safely for extended periods, protects components from overheating, and demonstrates the importance of cooling in electrical systems. The ultimate goal is to create a robust, safe, and efficient educational tool for labs and training.

4. Applications and Real-Time Use Cases

This testing board is a versatile tool with several practical applications:

- **Educational Laboratories:** Its primary use is in physics labs, vocational schools (ITIs), and engineering colleges for conducting experiments on Ohm's Law, Kirchhoff's Laws, and basic circuit properties.
- **Technical Training:** Ideal for training new electricians or technicians, providing a safe platform to practice circuit identification and troubleshooting.
- **Classroom Demonstrations:** Perfect for instructors to visually and quickly demonstrate the difference between series (like old holiday lights) and parallel (like home wiring) circuits.
- **Hobbyist Workbench:** A useful and safe test bench for DIY electronics hobbyists to test components or small circuit prototypes.

CHAPTER 3 - Literature Review

1. Overview of Existing Series–Parallel Testing Methods

A review of common electrical laboratory practices reveals that series-parallel circuit testing is predominantly conducted using two methods. The most common is the **manual re-wiring method**, which utilizes breadboards, patch panels, or simple terminal blocks. In this setup, students or technicians must physically disconnect and reconnect wires and components (like lamp holders) to change the circuit configuration from series to parallel. While this method teaches fundamental wiring skills, it is widely noted in educational literature as being time-consuming and highly susceptible to user error, leading to incorrect connections or potential short circuits.

The second method involves **pre-fabricated testing boards**. These range from simple, large-scale demonstration boards with knife switches to more complex, modular systems. Many existing boards use rotary switches or multiple toggle switches to select different circuit paths. However, a gap exists for a simple, cost-effective solution focused on a **single, direct comparison** (A/B testing) for one set of loads, which is the problem this project aims to solve.

2. Comparison of Manual vs. Modern Testing Boards

When comparing traditional, manual testing setups with modern, integrated testing boards, a clear trade-off emerges between flexibility and safety.

- **Manual Testing Boards** (e.g., breadboards, discrete components) offer maximum flexibility. Users can build infinitely complex and varied circuits. However, they lack inherent safety features. This approach exposes users to live terminals, increases the risk of component damage from incorrect wiring, and makes comparative testing (like quickly switching from series to parallel) a slow and cumbersome process.
- **Modern Testing Boards** (like the one proposed) prioritize **safety, speed, and reliability**. By enclosing all wiring, integrating fusing, and using a dedicated switch for configuration, the risk to the user is significantly minimized. The switching mechanism makes the comparison instantaneous, allowing the user to focus on observing the *effect*(e.g., change in lamp brightness) rather than on the *process* of re-wiring. This pedagogical advantage is a key justification for its development.

3. Role of Exhaust Fan in Heat Management

Thermal management is a critical but often overlooked aspect of electrical testing equipment. All electrical components, from wires to switches and especially loads (like incandescent bulbs), generate heat due to electrical resistance, governed by the principle of Joule heating ().

In an enclosed testing board, this heat can accumulate, leading to several problems:

1. **Component Derating:** Switches, wires, and insulation can degrade or fail prematurely when operated above their rated temperatures.
2. **Unreliable Results:** The resistance of components changes with temperature, which can skew the results of sensitive experiments.

The inclusion of an **exhaust fan** provides an **active cooling solution**. It works on the principle of forced convection, continuously expelling the hot, stagnant air from inside the enclosure and drawing in cooler, ambient air from the outside. This maintains a stable internal operating temperature, ensuring component longevity, reliable performance, and a safe-to-touch chassis.

CHAPTER 4 - System Requirement

1. Hardware Components

Fuse

- Provides safety to the circuit.
- It breaks the circuit during overload or short circuit.

Two-way Switch

- Used to control the mode of connection (Series or Parallel).
- Also used to turn ON/OFF the circuit.

Lamp Holder with Bulb

- Used to test electrical load in both series and parallel mode.
- Helps to observe brightness difference.

3-Pin Power Socket

- Used to test electrical appliances like chargers, iron box, mixer, etc.
- Provides phase (P), neutral (N), and earth.

Exhaust Fan

- Used for cooling and heat dissipation.
- Prevents overheating during continuous testing.

Connecting Wires

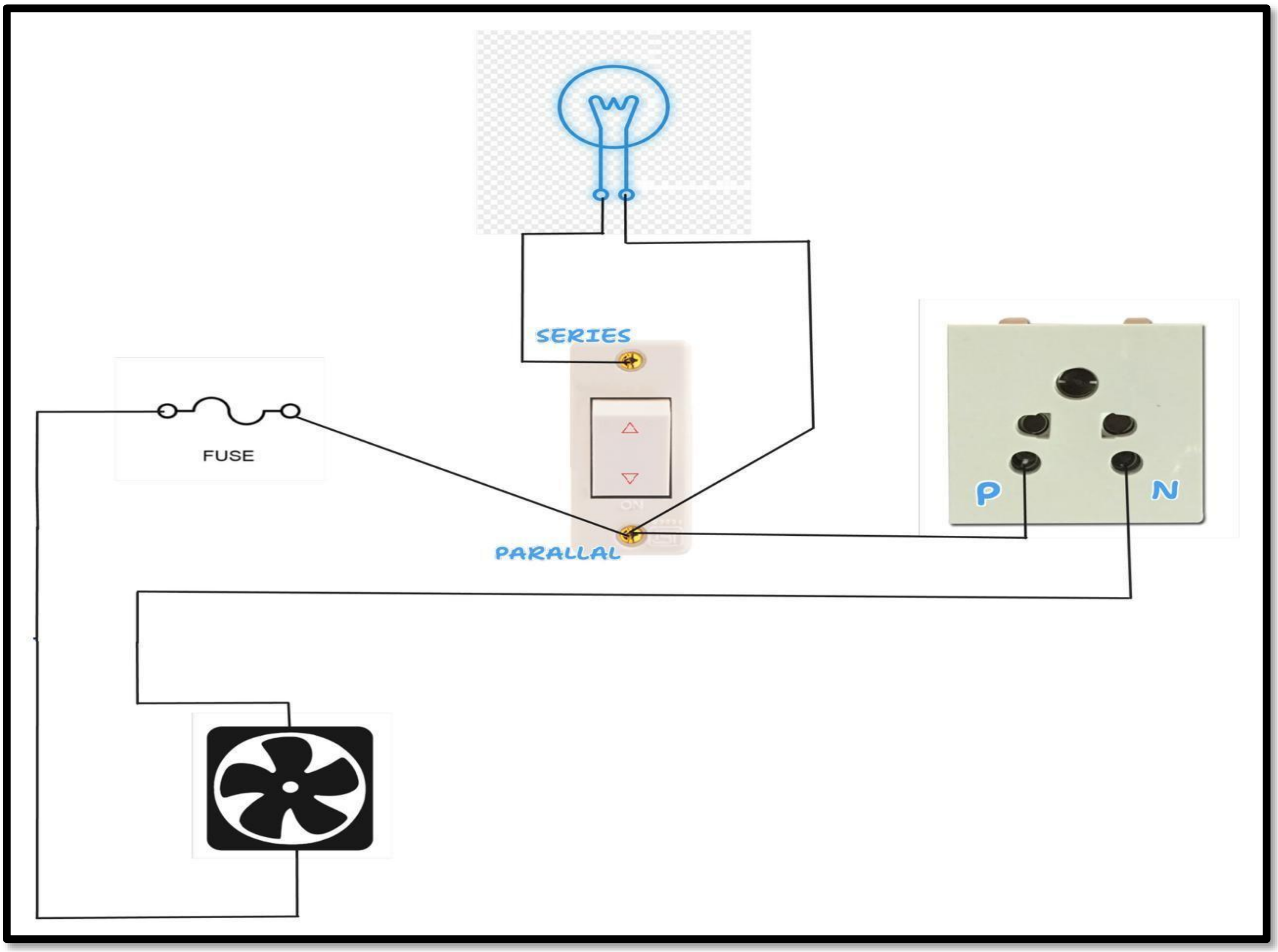
- Used for internal wiring.
- Phase and neutral are carefully routed to all components.

2. Software Tools

1. Design and Documentation Tools

- Circuit Design Software: A program like AutoCAD Electrical (for professional schematics) or Fritzing (for visual wiring layouts) used to design and finalize the circuit diagram before assembly.
- Project Documentation Software: A word processor such as Microsoft Word or Google Docs used to prepare the project report, compile observations, and integrate diagrams.

CHAPTER 5 - Circuit Diagram



5.0 Circuit Connection Explanation

The circuit starts from the main phase supply, which first passes through a fuse. The fuse is added for safety, so if there is any overload or short circuit, it will disconnect the supply and protect the circuit.

After the fuse, the phase wire is connected to the switch. This switch controls whether the load (bulb and socket) will work in series mode or parallel mode.

In series mode, the current flows through the switch to the bulb first, then it goes to the socket. Because of series connection, the voltage is shared, so the bulb will glow dim, and the connected appliance receives reduced voltage for safe testing.

In parallel mode, the switch connects the bulb and socket directly to the full phase voltage, so the bulb glows fully, and the socket gets full power. The neutral wire is directly given to the bulb, socket, and exhaust fan.

The exhaust fan is connected in parallel to the supply, so it always receives full voltage and works normally whenever the circuit is ON. Its purpose is to remove heat and provide ventilation during testing.

Thus, this circuit allows safe testing of appliances in both low-voltage (series) and full-voltage (parallel) modes with safety and cooling provided by the fuse and exhaust fan.

5.1 Power Flow Overview

The power flow in the circuit begins from the main AC supply (Phase and Neutral). The phase wire first passes through the fuse, which acts as a safety component. From the fuse, the phase is connected to the control switch, which decides whether the load will operate in series mode or parallel mode.

In series mode, the power flows from the switch to the bulb first, and then to the testing socket, providing reduced voltage for safe testing.

In parallel mode, the power flows directly from the switch to both the bulb and socket simultaneously, providing full voltage to both.

Meanwhile, the neutral wire is directly connected to all the components — bulb, socket, and exhaust fan — completing the circuit.

The exhaust fan remains connected in parallel with the main supply, so it always receives full power and helps to remove heat or smoke during testing.

Thus, the power is safely distributed and controlled between series and parallel paths as per the user's requirement.

2. Component-Wise Connection Details

1. Input Power Source

- The system receives power from a standard 3-pin AC mains plug.
- The Live (Phase) wire immediately connects to one end of the Fuse. This is the primary safety cut-off for the entire system.
- The Neutral wire connects to a common neutral junction, which feeds both the Exhaust Fan and the Neutral (N) terminal of the output socket.
- The Earth wire (if included) would be connected to the metal body of the enclosure (if any) and the Earth pin of the socket for safety.

2. Series Connection Path

- When the selector switch is in the "SERIES" position (rocker pressed up):
    1. The fused Live current flows to the center (common) terminal of the switch.
    2. The switch internally connects the common terminal to the top "SERIES" terminal.
    3. Power flows from this "SERIES" terminal, through the test bulb.
    4. After exiting the test bulb, the wire connects to the Phase (P) terminal of the output socket.
- **Result:** The test bulb and the appliance plugged into the socket are now connected in a series circuit.

5.2.3 Parallel Connection Path

- When the selector switch is in the "PARALLAL" (Parallel) position (rocker pressed down):
    1. The fused Live current flows to the center (common) terminal of the switch.
    2. The switch internally connects the common terminal to the bottom "PARALLAL" terminal.
    3. Power flows from this "PARALLAL" terminal directly to the Phase (P) terminal of the output socket.
- **Result:** The test bulb is completely bypassed. The appliance in the socket receives the full, direct mains voltage in parallel.

5.2.4 Exhaust Fan Wiring

- The exhaust fan is wired directly in parallel with the main power input (after the fuse).
- One wire from the fan connects to the fused Live line.
- The other wire from the fan connects to the main Neutral line.
- **Result:** The fan operates at all times, independent of the switch position, to provide constant cooling as long as the board is powered on.

5.2.5 Output Load/Test Points

- The Output Socket (e.g., 16A 6-pin socket) serves as the single test point for connecting the external load or appliance.
- Its Phase (P) terminal receives the switched power (either via the series bulb or directly).
- Its Neutral (N) terminal provides the return path, connecting directly to the main Neutral line.

CHAPTER 6 - Working Principle

1. Electrical Flow Overview

This project operates on the principle of controlling the voltage supplied to the load by using series and parallel connections.

When the power supply is turned ON, the phase first passes through a fuse, which acts as a safety device against overcurrent. From the fuse, the phase is given to a control switch, which allows the user to select either series mode or parallel mode for testing the electrical load.

- **Series Mode:**
  In this mode, the phase passes through the bulb first and then goes to the testing socket. Since the bulb and socket are connected in series, the voltage is divided, and the current flow is limited. As a result, the bulb glows dim, and the connected appliance receives reduced voltage, making it safe for initial testing or fault detection.
- **Parallel Mode:**
  In this mode, the switch directly connects both the bulb and testing socket to the full phase supply in parallel. Both receive full voltage independently, so the bulb glows at full brightness, and the connected appliance operates under normal operating conditions.

The neutral line is common and directly connected to all components (bulb, socket, and exhaust fan), completing the return path of the circuit.

An exhaust fan is permanently connected in parallel to the main supply so that it runs continuously when the board is in use. It helps to remove heat and provide ventilation, ensuring safe testing conditions.

2. Step-by-Step Process of Testing

6.2.1Preparation & Safety Checks

1. **Wear PPE** — insulated gloves, safety goggles, no wet hands or jewelry.
2. **Ensure mains isolation** — switch OFF main supply before making connections.
3. **Check wiring & earthing** — verify all wires are correctly connected (Phase, Neutral, Earth). Ensure earth is continuous.
4. **Verify components** — fuse rating corrects and intact, bulb fitted, socket tight, fan secured.
5. **Instruments ready** — keep a multimeter (voltmeter & ammeter), clamp meter (optional), screwdriver, and observation sheet ready.
6. **Emergency plan** — know where the main switch and fire extinguisher are.

6.2.2 Initial No-Load Check

1. Turn ON the main supply (with no appliance in the socket).
2. Check exhaust fa**n** runs (it is permanently in parallel). If it doesn't, switch OFF and inspect wiring.
3. With multimeter, measure line-to-neutral voltage at the board (should read mains ~230 V AC). Record value.

3. Testing in Series Mode

1. Place the control switch to Series position.
2. Insert the test appliance plug into the 3-pin socket (or leave socket empty to observe bulb effect first).
3. Turn ON the board switch. Observe the bulb:
    - It should glow dim (because voltage divides in series).
4. Measurements (recommended):
    - Voltage across bulb — measure and record.
    - Voltage at socket terminals — measure and record (should be less than mains in series).
    - Current through circuit — measure (same through bulb and appliance in series).
5. Observe appliance behaviour:
    - Appliance should run weakly or not at full power (safe for initial testing).
6. Record observations: bulb brightness, voltages, current, any unusual noise/heating/smell.

4. Testing in Parallel Mode

1. Turn OFF the board. Move the switch to Parallel position.
2. Insert or keep the same appliance in socket and turn ON board.
3. **Observations:**
    - Bulb should glow full brightness.
    - Appliance should operate normally at full power.
4. **Measurements:**
    - Voltage at socket — should measure mains voltage (approx. same as initial line-to-neutral).
    - Current drawn by appliance — record.
    - Voltage across bulb — record (should be mains).
5. Record any heating, sparking, abnormal noise.

5. Exhaust Fan Check

- Verify fan runs at full speed during both modes (since it is in parallel).
- Observe that it effectively removes heat/smoke if appliance heats up.

6.2.6 Fault / Overload Simulation (Optional & only if instructed)

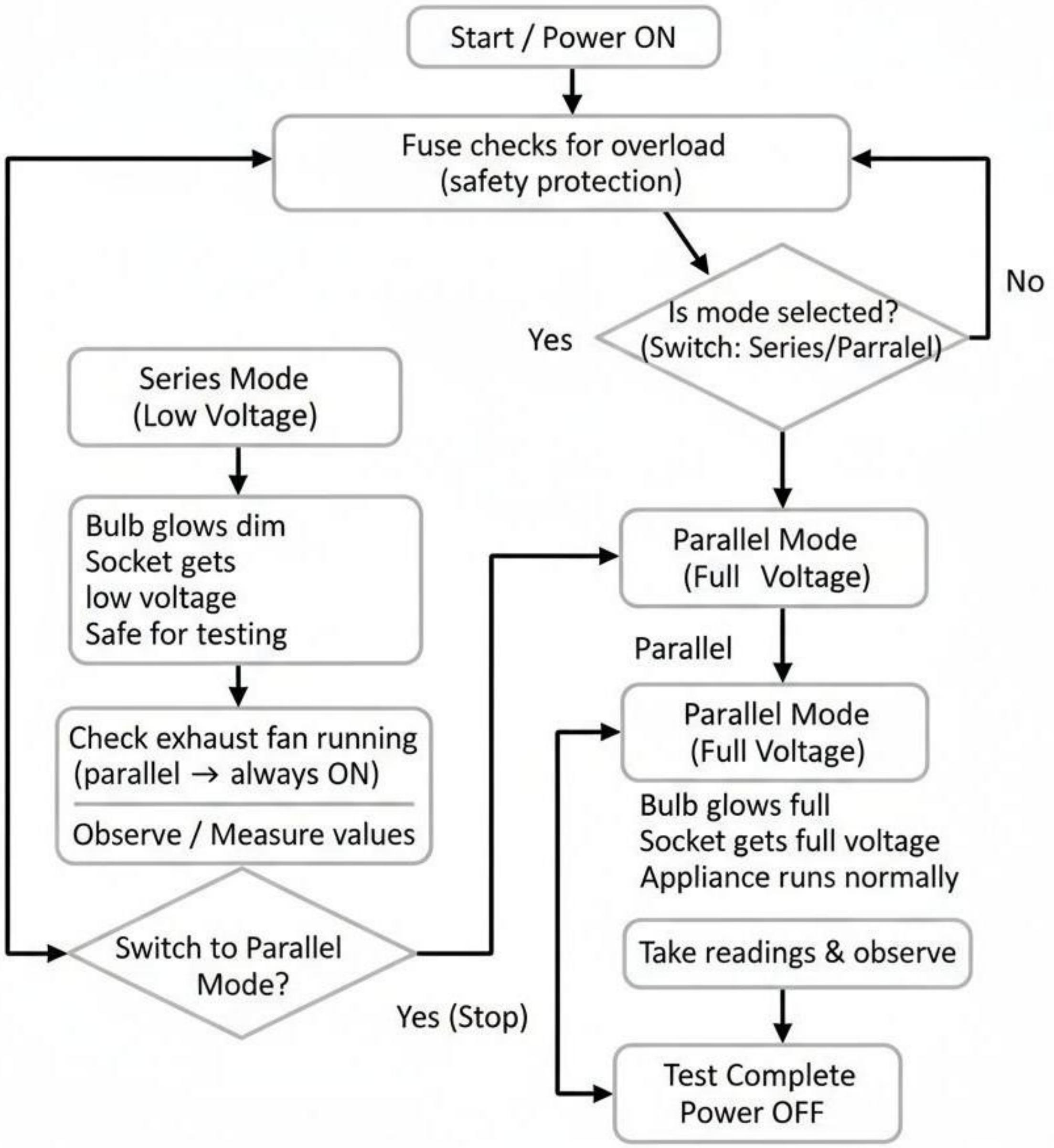
Only perform if supervised and safe — do not deliberately short circuit.

1. With appropriate precautions, simulate a small overload by connecting a known load and monitor fuse response.
2. Confirm fuse blows / trips when current exceeds fuse rating — verify protection works.

6.2.7 Shutdown & Final Checks

1. Turn OFF the board and unplug the appliance.
2. Inspect components for heating, discoloration, loose screws.

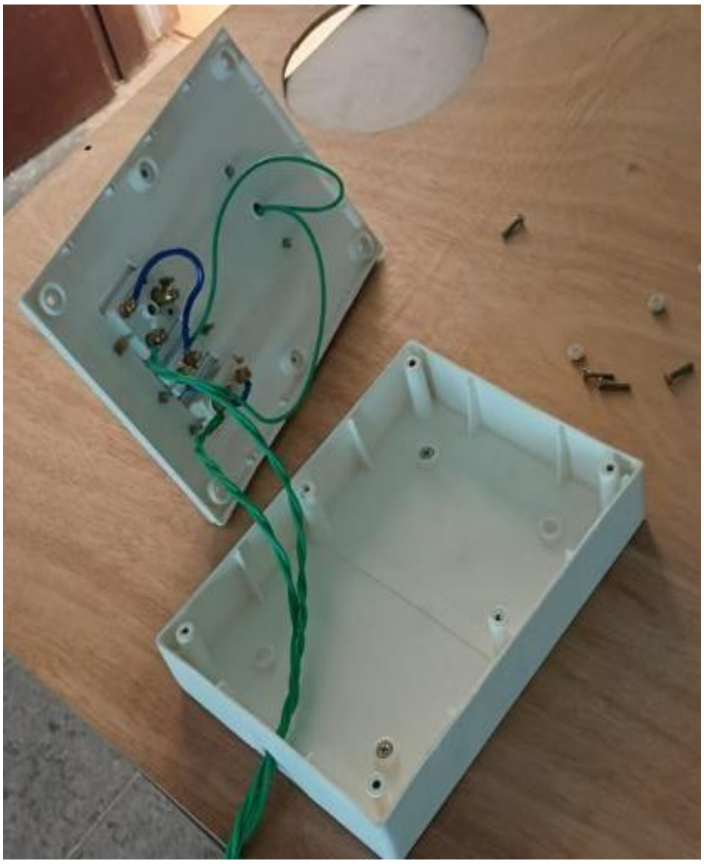
Flowchart:

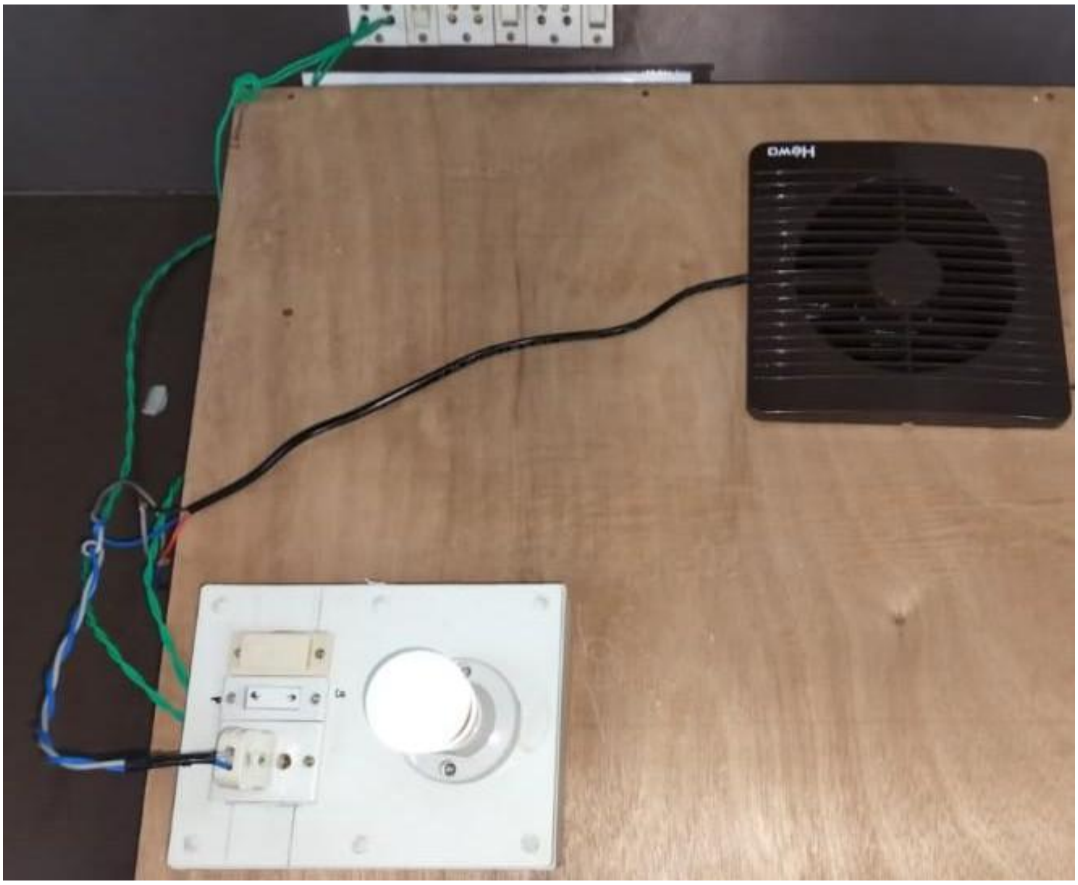

Start / Power ON
Fuse checks for overload
(safety protection)
Is mode selected?
(Switch: Series/Parralel)
Yes
No
Series Mode
(Low Voltage)
Bulb glows dim
Socket gets
low voltage
Safe for testing
Check exhaust fan running
(parallel → always ON)
Observe / Measure values
Switch to Parallel
Mode?
Yes (Stop)
Parallel Mode
(Full Voltage)
Parallel
Parallel Mode
(Full Voltage)
Bulb glows full
Socket gets full voltage
Appliance runs normally
Take readings & observe
Test Complete
Power OFF

CHAPTER 7 : Testing and Results

7.1 Hardware Testing Procedures

Step 1: Inspection
Verified correct wiring of Phase, Neutral, and Earth. Ensured fuse, bulb, socket, and exhaust fan were properly connected.
Step 2: Power-On Test
Switched ON the supply — exhaust fan started immediately. No signs of spark, smell, or noise.
Step 3: Series Mode Test
Switched to Series mode — bulb glowed dim and appliance received low voltage for safe testing.
Step 4: Parallel Mode Test
Switched to Parallel mode — bulb glowed fully and appliance ran at normal 230V. Fan stayed ON.
Step 5: Measurement (Optional)
Measured voltage and current using multimeter — readings were correct.
Step 6: Shutdown
Switched OFF the power and confirmed no overheating or loose connections
Testing successful.





CHAPTER 8 – CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project “1 Point Series–Parallel Testing Board with Exhaust Fan” has been successfully designed, implemented, and tested to demonstrate the principles of electrical connections and appliance testing. The main objective of this setup was to create a safe and efficient testing board that can operate electrical loads in both series and parallel modes, allowing users to study voltage, current, and power behaviour in each case.

In series mode, the connected load receives a reduced voltage, making it ideal for safe preliminary testing of electrical appliances. In parallel mode, the load receives the full supply voltage, enabling normal operation. The fuse provides necessary protection against overload or short circuit, while the exhaust fan, connected in parallel, ensures continuous cooling and proper ventilation, preventing overheating of components during operation.

This project helps in understanding the practical difference between series and parallel circuits, as well as the importance of safety components like fuses, earthing, and controlled switching. It is highly useful for educational laboratories, workshops, and basic electrical testing setups, where safe and reliable operation is essential.

8.2 Future Enhancements

Although the present “1 Point Series–Parallel Testing Board with Exhaust Fan” performs its intended function efficiently, there is scope for several improvements to make it more advanced, user-friendly, and safer.

Digital Measurement System:

Integration of digital voltmeters and ammeters can display real-time voltage, current, and power readings for better monitoring and analysis.

Indicator Lamps or LEDs:

Adding LED indicators to show series or parallel mode selection, power ON/OFF status, and fault conditions will enhance user convenience and safety.

Overload & Short-Circuit Alarm:

A buzzer or alarm system can be added to alert the user in case of overload or short-circuit conditions.

Automatic Mode Selection:

Incorporating a microcontroller-based system can automatically switch between series and parallel modes depending on the appliance requirement.

8.3 References:

Textbooks & Manuals:

- B.L. Theraja & A.K. Theraja, *“A Textbook of Electrical Technology”*, S. Chand Publications

Web References:

- https://www.electrical4u.com — Concepts of series and parallel circuits.
- https://www.electronics-tutorials.ws — Electrical wiring and safety practices.

ASSESSMENT

Internal:

SL NO	RUBRICS	FULL MARKS	MARKS OBTAINED	REMARKS
1	Understanding the relevance, scope and dimension of the project	10		
2	Methodology	10		
3	Quality of Analysis and Results	10		
4	Interpretations and Conclusions	10		
5	Report	10		
	Total	**50**		

Date: **Signature of the Faculty**

COURSE OUTCOME (COs) ATTAINMENT

Expected Course Outcomes (COs):

(Refer to COs Statement in the Syllabus)

➢ **Course Outcome Attained:**

How would you rate your learning of the subject based on the specified COs?

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

1 2 3 4 5 6 7 8 9 10

LOW **HIGH**

➢ **Learning Gap (if any):**

➢ **Books / Manuals Referred:**

Date: **Signature of the Student**

➢ **Suggestions / Recommendations:**

(By the Course Faculty)

Date: **Signature of the Faculty**

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